

JURONG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

CLASS

BIOLOGY

9648/03

Applications Paper and Planning Question

14 September 2015

Paper 3

2 hours

Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	
2	
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5	
Total	

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Answer **all** the questions in this section.

- 1** Human insulin can be synthesised in a laboratory strain of *E. coli* using recombinant DNA (rDNA) technology. The starting point for the process is mature insulin mRNA, isolated from human pancreatic cells.

(a) Explain what is meant by *recombinant DNA (rDNA)*. [2]

(b) State the role of each of these enzymes in producing rDNA carrying the insulin gene. [2]

(i) reverse transcriptase

(ii) restriction enzyme

(c) A DNA library is a collection of cloned DNA fragments. The two types of DNA libraries, genomic library and cDNA library, allows one to isolate a specific gene for use in rDNA technology.

Explain why insulin gene is isolated from a cDNA library instead of a genomic DNA library. [2]

Plasmid vectors such as pGLO are used to introduce insulin genes into *E. coli* by transformation and enable replication of these insulin genes.

In the synthesis of pGLO, the genes that code for arabinose catabolism (*araB*, *araA*, and *araD*) have been replaced by the green fluorescent protein (GFP) gene. Colonies appear fluorescent green under UV light when GFP gene is expressed and white when GFP gene is not expressed. The multiple cloning sites is found within the GFP gene. In the presence of arabinose, the arabinose interacts directly with AraC protein. AraC protein change shape and promotes the binding of RNA polymerase to P_{BAD}.

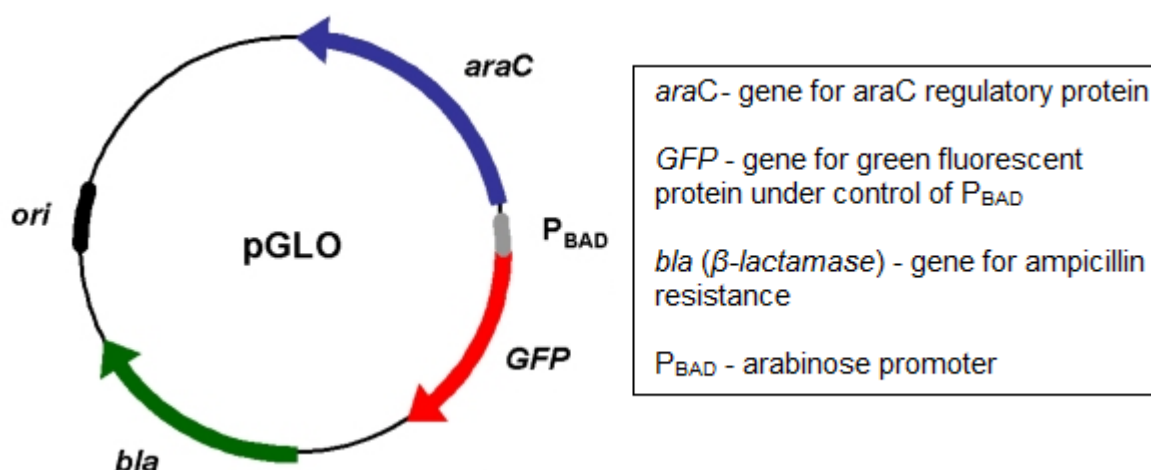


Fig. 1.1

(d) With reference to pGLO shown in Fig. 1.1, describe two properties of plasmids that allow them to be used as cloning vectors. [4]

(e) After the transformation process, bacterial suspensions are spread onto plates X and Y.

(i) Complete Table 1.1 to show the observations made of each plate when exposed to UV light. [2]

Table 1.1

plate	content	observation(s)
X	nutrient rich media / Amp	
Y	nutrient rich media / Amp / arabinose	

(ii) With reference to your observations in Table 1.1, explain which plate(s) may contain colonies that have taken up the insulin gene. [3]

[Total: 15]

- 2 Cystic fibrosis (CF) is caused by mutations within the gene encoding the cystic fibrosis transmembrane conductance regulator (CFTR) protein. The most common mutation in this gene results in the loss of one amino acid from the CFTR protein. This deletion, $\Delta F508$, accounts for about 70% of mutations in cystic fibrosis.

(a) (i) State the type of mutation responsible for this change in the amino acid sequence. [1]

(ii) Outline the role of the product of the normal *CFTR* allele. [3]

Since the *CFTR* gene was discovered and cloned, gene therapy to introduce a normal copy of the gene has been a possibility for CF treatment.

In 2012, the UK Cystic Fibrosis Gene Therapy Consortium launched a trial involving 140 patients to assess the clinical efficacy of a non-viral CFTR gene-liposome complex as a delivery system for a functional CFTR gene. 78 of whom received at least 9 monthly doses of the therapy, while a control group of 62 inhaled saline solution. The results from the year-long trial, showed that those who had been breathing in the gene treatment had, on average, a 3.7 percent improvement in their lung function compared with the placebo group.

Over the course of the year, the lung function of patients in the treatment group seemed to stabilise, while the control group showed a steady decline in lung power.

(b) (i) Outline how CFTR gene-liposome complex can be used as part of a gene therapy treatment for CF. [3]

(ii) Account for the 3.7 percent improvement in the patients' lung function. [1]

In 2005, a group of scientist explored using adult stem cells from bone marrow, referred to as marrow stromal stem cells (MSCs), to provide a therapy for Cystic Fibrosis (CF). They found that MSCs can be induced to differentiate into airway epithelial cells.

The MSCs are amenable to *CFTR* gene correction using adenoviral vector. This serves as another attractive potential method of correcting the basic defect in respiratory epithelium.

(c) (i) Describe the features of MSCs and their role in CF gene therapy. [3]

(ii) Explain the advantages of using an adenoviral vector as the gene delivery system for the normal allele of the *CFTR* gene. [2]

[Total: 13]

- 3 Bananas are the second largest food-fruit crops of the world produced in the tropical and subtropical regions of mostly the developing countries. Bananas are important as a crop but the planting material required is often in short supply. Scientists have developed a micropropagation system as shown in Fig. 3.1.

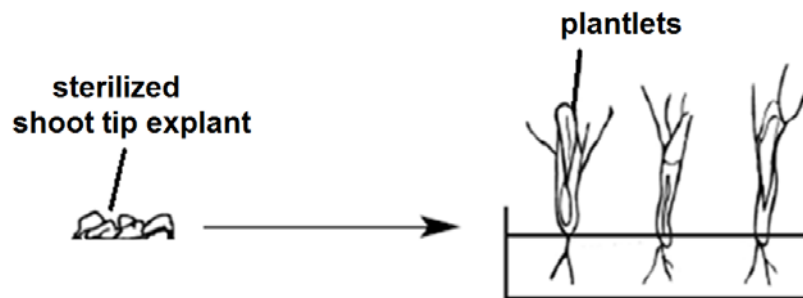


Fig. 3.1

- (a) With reference to Fig. 3.1, outline the techniques of cloning plants from tissue culture. [3]

- (b) State two environmental conditions that need to be controlled during the development of the plantlets. [1]

- (c) Suggest two advantages of propagating banana plants using this system. [2]

The effect of plant growth regulators, IAA and Kinetin, on shoot regeneration is shown in Table 3.1.

Table 3.1

MS medium + growth regulators (mg l^{-1})		Explants differentiated	Mean number of shoots per explant
+ IAA	+Kinetin		
0	0	29	2
0.1	1.0	42	4
0.1	2.5	46	8
0.1	5.0	58	8
0.5	1.0	54	12
0.5	2.5	66	20
0.5	5.0	58	28

(d) With reference to Table. 3.1,

(i) state the evidence that growth regulators benefit propagation, [1]

(ii) explain why a growth regulator combination of 0.5 mg l^{-1} IAA + 5.0 mg l^{-1} kinetin is recommended. [1]

Fig. 3.2 shows the proportion of cotton, maize and soybeans grown in the USA that are genetically modified in two different ways:

- HT crops are modified to be resistant to the glyphosate herbicide.
- Bt crops are modified to express the Bt toxin which kills insect pests.

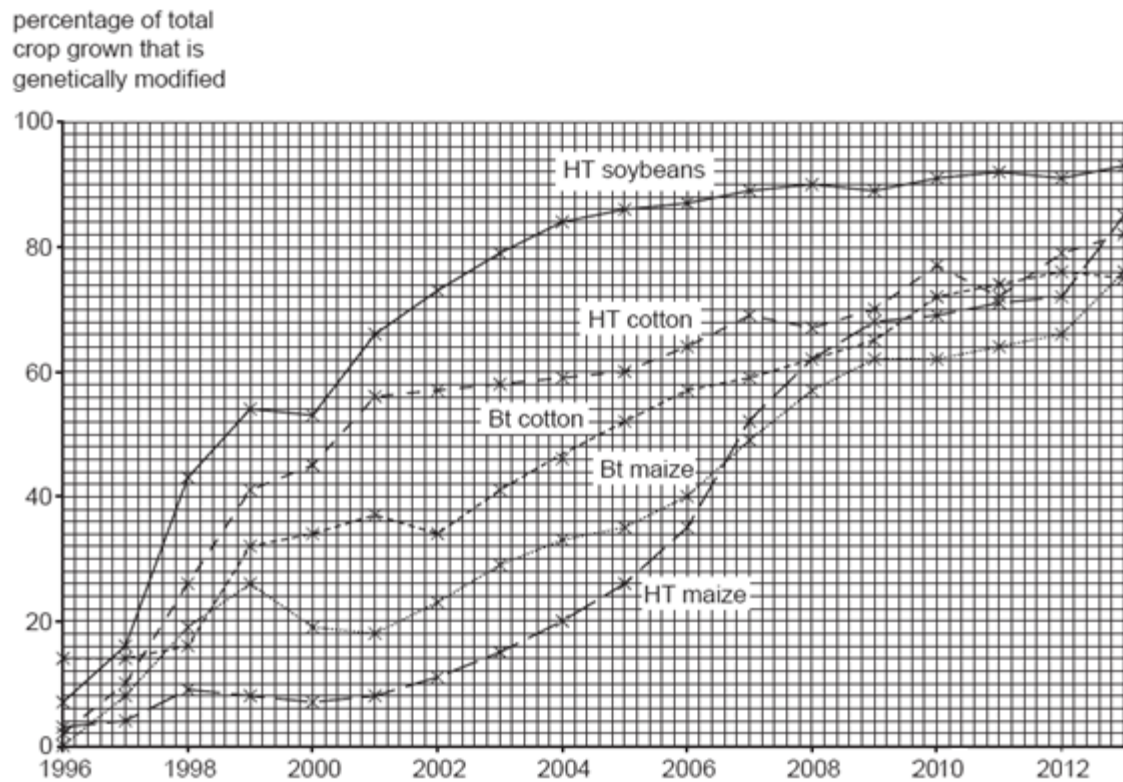


Fig. 3.2

(e) (i) With reference to Fig. 3.2, describe the change in the percentage of genetically modified soybeans grown in the USA since 1996. [2]

Fig. 3.3 shows the increase in the number of weed species resistant to glyphosate herbicide and triazine herbicide since 1970.

Crops have not been genetically modified to resist triazine herbicide.

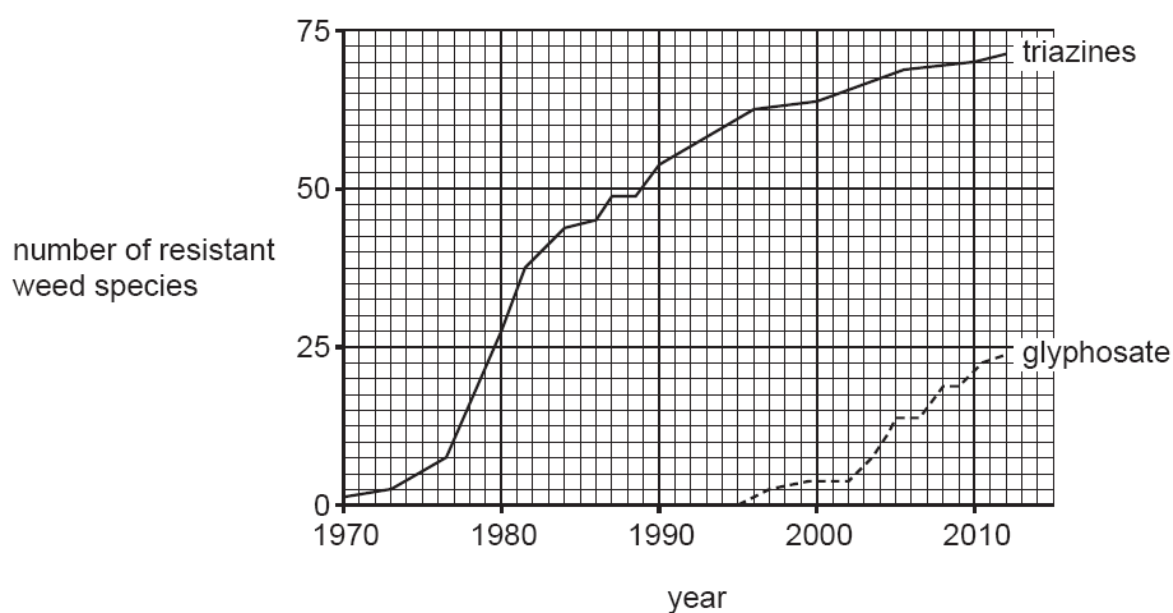


Fig. 3.3

- (ii) With reference to Fig. 3.2 and Fig. 3.3, suggest if the rise in glyphosate-resistant weeds has resulted from the introduction of genetically modified crops. [1]

- (iii) Suggest one possible risk to the environment as a result of growing genetically modified crops. [1]

[Total: 12]

4 Planning Question

The green algae, *Scenedesmus quadricauda*, contain the photosynthetic pigments chlorophyll a and b. This enables the green algae to undergo photosynthesis.

S. quadricauda require carbon dioxide that is dissolved in water to photosynthesize. Water does not naturally contain a lot of dissolved carbon dioxide. When sodium hydrogen carbonate is dissolved in water, it produces carbon dioxide and increases the amount of carbon dioxide in the water. The concentration of hydrogen carbonate ions in a solution is directly proportional to the concentration of carbon dioxide available.

S. quadricauda can be immobilised in alginate beads. The alginate beads are to be arranged in a monolayer to fully cover the bottom of a conical flask.

Using this information and your own knowledge, design an experiment to test the hypothesis that:

“The rate of photosynthesis is dependent on the concentration of carbon dioxide”

You must use:

- an unlimited supply of *S. quadricauda* alginate beads all of diameter 11 mm,
- an unlimited volume of 1.0 % sodium hydrogen carbonate,
- an unlimited volume of distilled water,
- 250 cm³ conical flasks of diameter 85 mm,
- 100 cm³ gas syringe.

You may select from the following apparatus:

- any normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, glass rod, etc.,
- syringes,
- 100-watt table lamp,
- timer e.g. stopwatch.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- be illustrated by relevant diagram(s), if necessary,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- include layout of results tables and graphs with clear headings and labels,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

[illegible]

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[illegible]

5 Free-response question

Write your answers to this question on the separate answer paper provided.

Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate.
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

(a) Describe the polymerase chain reaction (PCR) and explain how PCR facilitates genetic fingerprinting. [8]

(b) Explain how RFLP analysis facilitated the process of DNA fingerprinting. [6]

(c) Discuss the ethical concerns that have arisen about the human genome project. [6]

[Total: 20]